Process Synchronization using xv6-public

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1. Setup

A kernel-managed shared page is mapped through shmget(). Kernel semaphores use xv6 sleep()/wakeup() and a spinlock.

2. Peterson

Two processes use flag[2] and turn to implement mutual exclusion without an OS lock. Final counter: 20; violations: 0.

3. Producer-consumer

A five-slot circular buffer uses empty, full, and mutex semaphores. Twenty values are delivered FIFO.

4. Readers-writers

Three readers and two writers share a protected integer. Concurrent reads and exclusive writes are verified. Final value: 8; violations: 0.

5. Dining philosophers

Five fork semaphores are acquired in increasing resource order, preventing circular wait. All philosophers complete five cycles.

6. Results

Peterson PASS; PRODCONS PASS; READWRITE PASS; DINING PASS. The diffs, tests, and screenshots provide reproducible evidence.

Question	Result
Q1 Peterson	Counter 20; violations 0
Q2 Producer-consumer	20 produced; 20 consumed FIFO
Q3 Readers-writers	Final value 8; violations 0
Q4 Dining philosophers	All philosophers complete 5 cycles